CH10. Fundamentals of the Metal-Oxide-Semiconductor Field-Effect Transistor (Part I)

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10장 용어 정리

축적층 전하 *accumulation layer charge*	열평형 다수 캐리어 농도보다 큰, 산화막 바로 아래에 유기된 전하
채널 컨덕턴스 *channel conductance*	$V_{DS} \rightarrow 0$ 조건에서 드레인-소스 전압 변화에 대한 드레인 전류의 변화량
채널 컨덕턴스 변조 *channel conductance modulation*	채널 컨덕턴스가 게이트-소스 전압에 따라 변하는 과정
CMOS (상보형 MOS) *complementary MOS*	단일 반도체 칩의 회로에서 p채널과 n채널 소자를 함께 이용하는 기술
전도 파라미터 *conduction parameter*	MOSFET 전류-전압 관계식에 곱해지는 계수
차단 주파수 *cutoff frequency*	ac 입력 게이트 전류와 ac 출력 드레인 전류가 같아지는 신호 주파수
공핍형 MOSFET *depletion mode MOSFET*	소자를 차단하려면 게이트 전압을 인가해야 하는 MOSFET
증가형 MOSFET *enhancement mode MOSFET*	소자를 도통시키려면 게이트 전압을 인가해야 하는 MOSFET
등가 고정 산화막 전하 *equivalent fixed oxide charge*	산화막-반도체 계면에 인접한 산화막 내의 유효 고정 전하 Q'_{ss}
전계 효과 *field-effect*	반도체 표면에 수직인 전계로 반도체의 전도도를 변화시키는 현상
평탄대 전압 *flat-band voltage*	산화막 아래 반도체에 공간전하 영역이 없는 평탄대 상태를 만드는 게이트 전압
계면 상태 *interface states*	산화막-반도체 계면에서 금지대 안에 허용되는 전자 에너지 상태
반전층 전하 *inversion layer charge*	산화막 바로 아래에 유기된, 반도체 도핑과 반대 유형의 전하
반전층 이동도 *inversion layer mobility*	반전층 내 캐리어의 이동도
금속-반도체 일함수 차 *metal-semiconductor work function difference*	금속 일함수와 반도체 전자친화도의 함수인 파라미터 ϕ_{ms}
산화막 커패시턴스 *oxide capacitance*	산화막 유전율과 두께의 비로 정한 단위 면적당 커패시턴스 C_{ox}

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10.1 The Two-Terminal MOS structure

10.2 Capacitance-Voltage Characteristics

10.3 The Basic MOSFET Operation

MOS Structure

Metal – Oxide (SiO_2) – Semiconductor (Si)

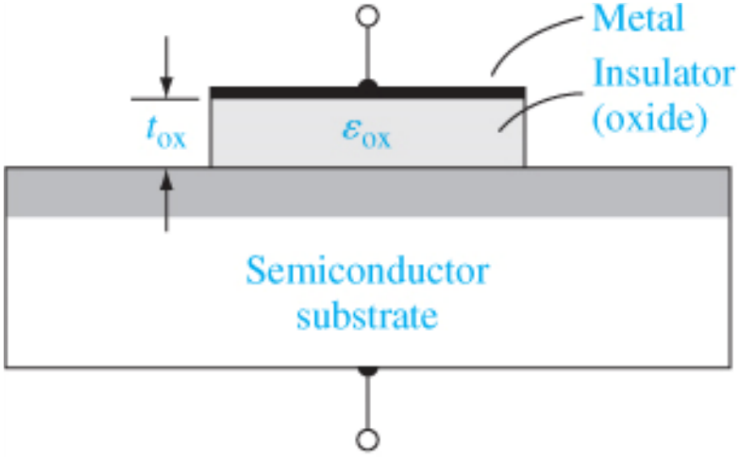


Figure 10.1 | The basic MOS capacitor structure.

10.1.1 Energy Band Diagrams

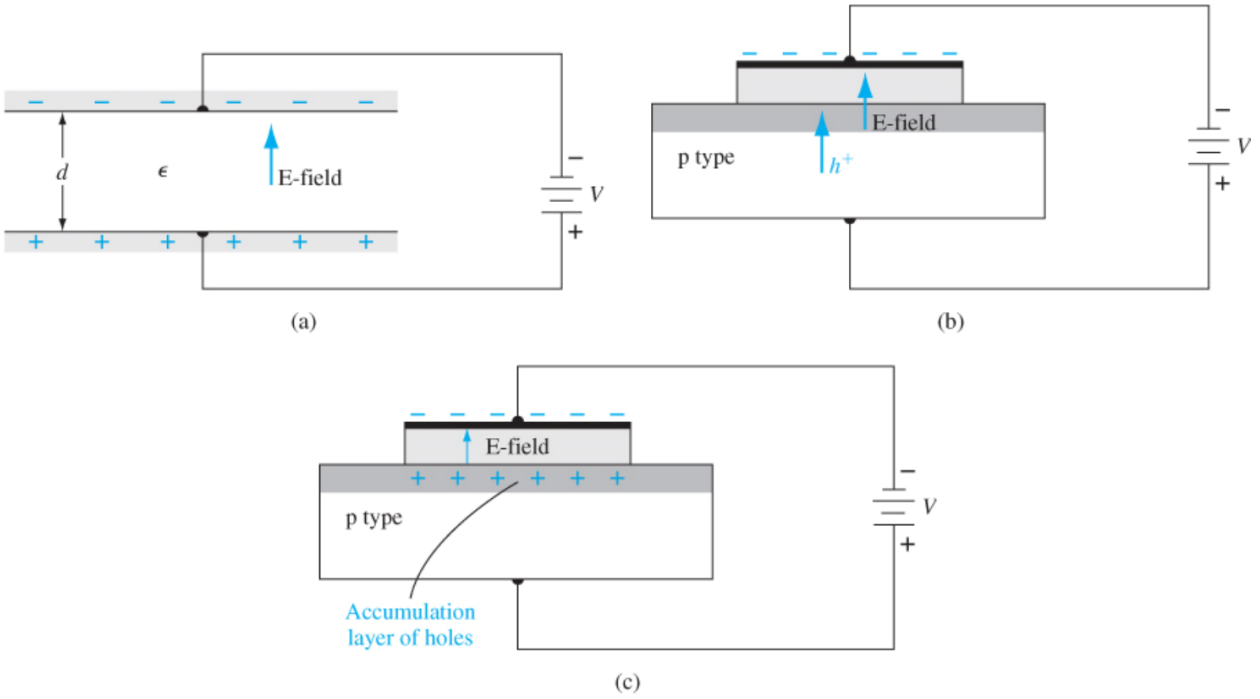


Figure 10.2 | (a) A parallel-plate capacitor showing the electric field and conductor charges. (b) A corresponding MOS capacitor with a negative gate bias showing the electric field and charge flow. (c) The MOS capacitor with an accumulation layer of holes.

10.1.1 Energy Band Diagrams

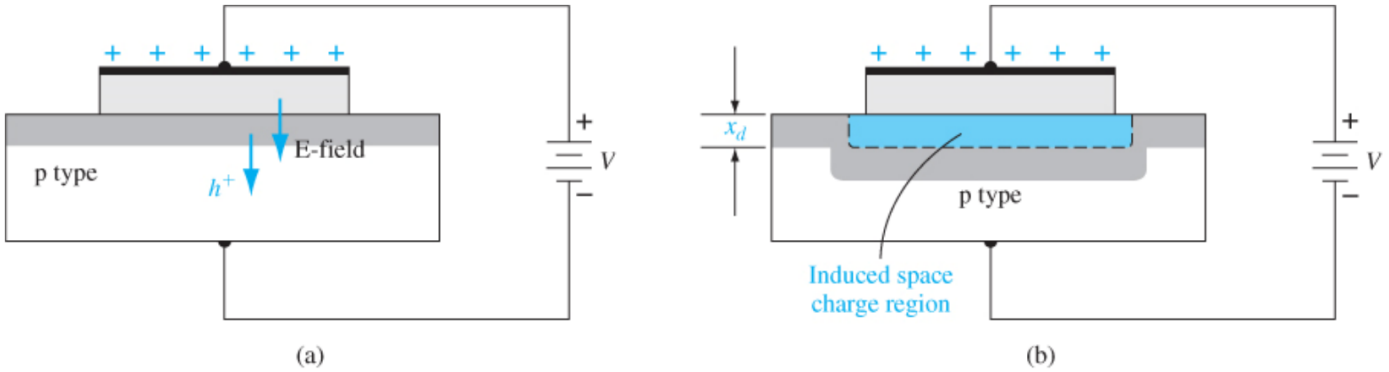


Figure 10.3 | The MOS capacitor with a moderate positive gate bias, showing (a) the electric field and charge flow and (b) the induced space charge region.

10.1.1 Energy Band Diagrams

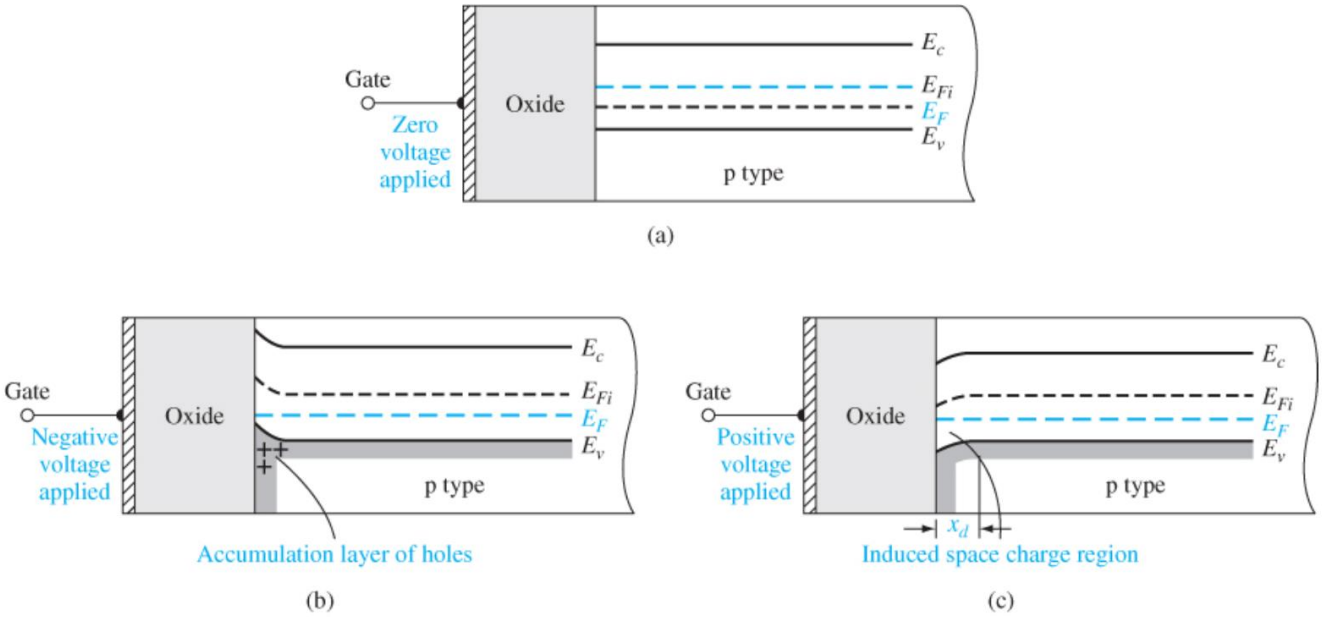


Figure 10.4 | The energy-band diagram of a MOS capacitor with a p-type substrate for (a) a zero applied gate bias showing the *ideal* case, (b) a negative gate bias, and (c) a moderate positive gate bias.

10.1.1 Energy Band Diagrams

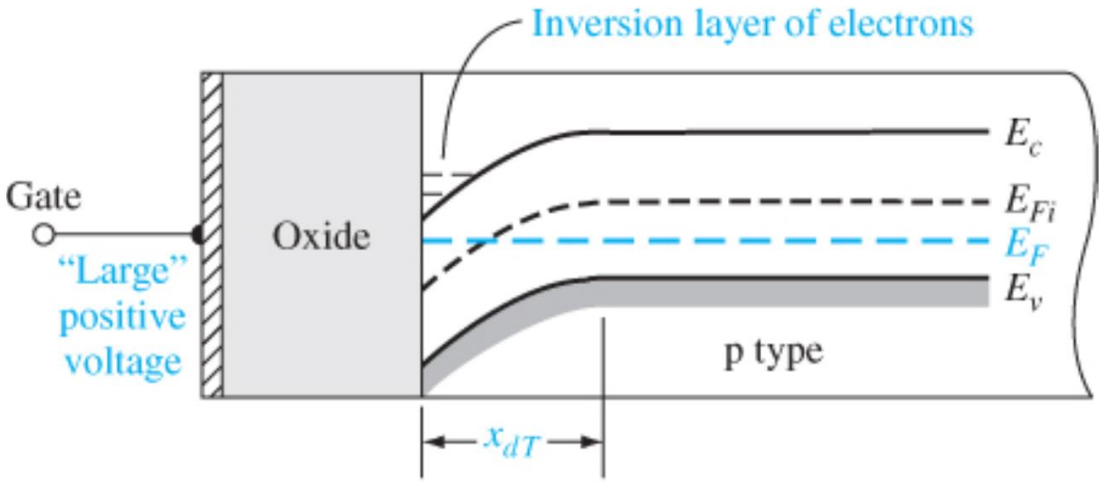


Figure 10.5 | The energy-band diagram of the MOS capacitor with a p-type substrate for a "large" positive gate bias.

10.1.1 Energy Band Diagrams

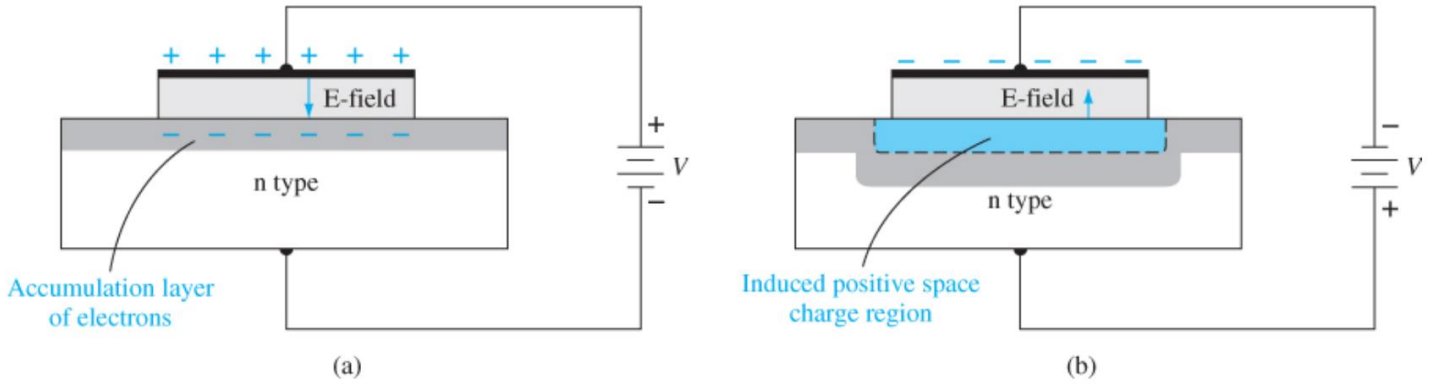


Figure 10.6 | The MOS capacitor with an n-type substrate for (a) a positive gate bias and (b) a moderate negative gate bias.

10.1.2 Depletion Layer Thickness

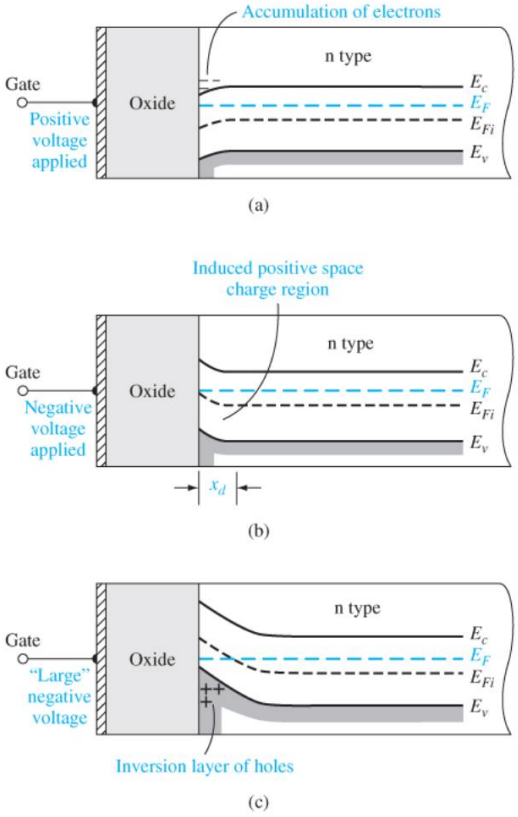


Figure 10.7 | The energy-band diagram of the MOS capacitor with an n-type substrate for (a) a positive gate bias, (b) a moderate negative bias, and (c) a "large" negative gate bias.

$$\phi_{fp} = V_t In\left(\frac{N_a}{n_i}\right) \quad (10.4)$$

$$x_d = \left(\frac{2\varepsilon_s \phi_s}{eN_a}\right) \quad (10.5)$$

10.1.2 Depletion Layer Thickness

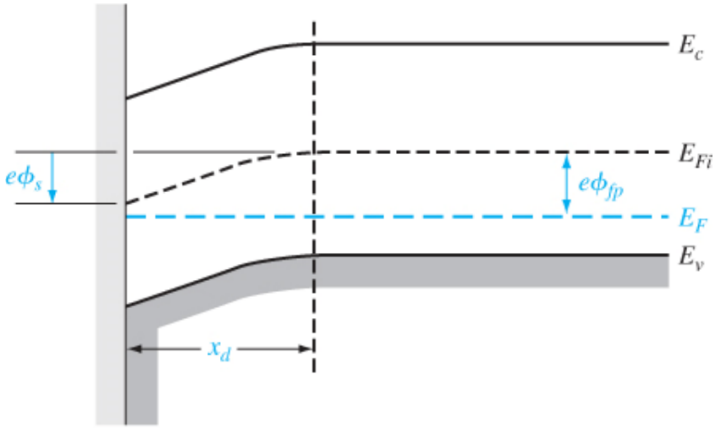


Figure 10.8 | The energy-band diagram in the p-type semiconductor, indicating surface potential.

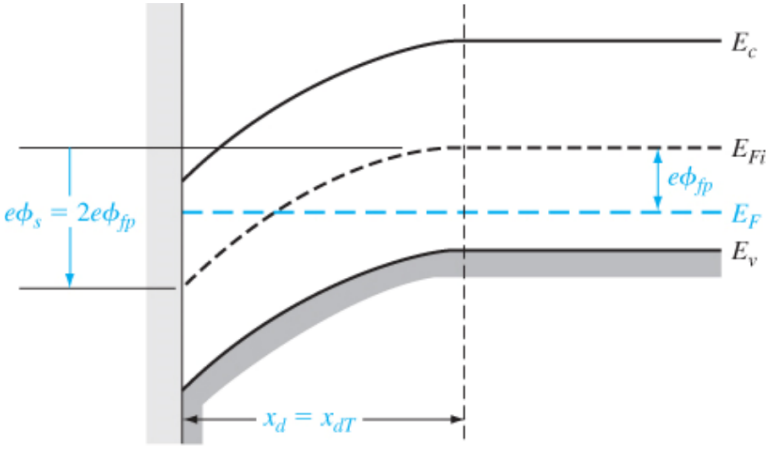


Figure 10.9 | The energy-band diagram in the p-type semiconductor at the threshold inversion point.

$$x_{dT} = \left(\frac{4\varepsilon_s \phi_{fp}}{eN_a} \right)^{1/2} \qquad (10.6)$$

Example 10.1

① **Calculate the maximum space charge width for a given semiconductor doping concentration**

- **T=300K, $N_a=10^{16}$ cm^{-3}**
- **The intrinsic carrier concentration is $n_i=1.5\times10^{16}$ cm^{-3}**

From Equation (10.4), we have

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right) = (0.0259)\ln\left(\frac{10^{16}}{1.5\times10^{10}}\right) = 0.3473V$$

Then the maximum space charge width is

$$\chi_{dT} = \left[\frac{4\varepsilon_s\phi_{fp}}{eN_a}\right]^{1/2} = \left[\frac{4(11.7)(8.85\times10^{-14})(0.3473)}{(1.6\times10^{-19})(10^{16})}\right]^{1/2}, or \quad \chi_{dT} \cong 0.30\times10^{-4}cm = 0.30\mu m$$

- **The maximum induced space charge width is on the same order of magnitude as pn junction space charge widths.**

10.1.2 Depletion Layer Thickness

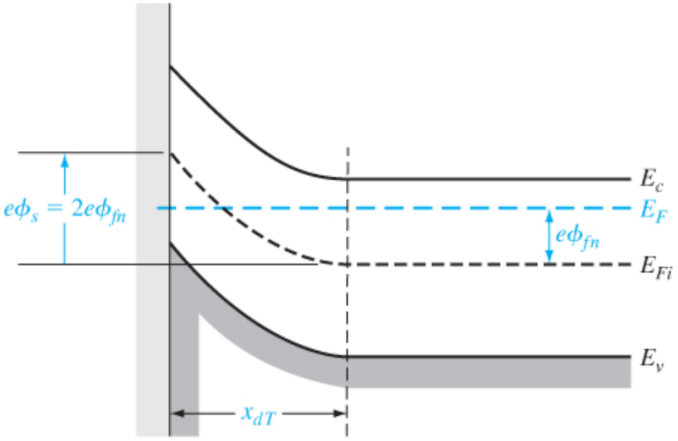


Figure 10.10 | The energy-band diagram in the n-type semiconductor at the threshold inversion point.

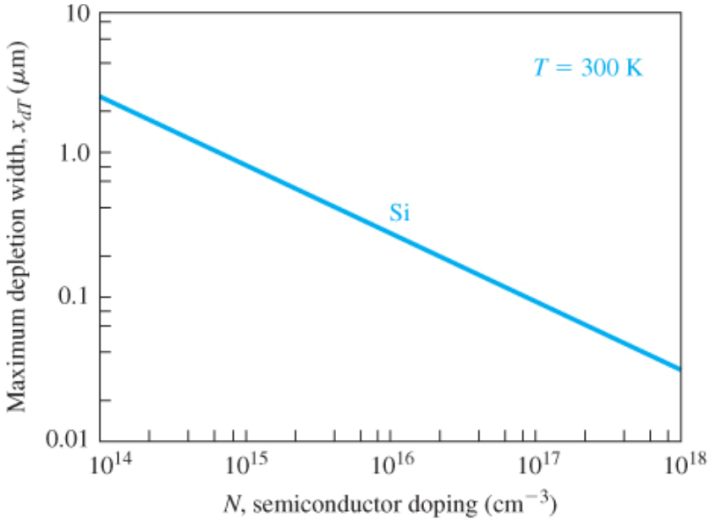


Figure 10.11 | Maximum induced space charge region width versus semiconductor doping.

$$\phi_{fn} = V_t In\left(\frac{N_d}{n_i}\right) \quad (10.7), \; x_{dT} = \left(\frac{4\varepsilon_s \phi_{fn}}{eN_d}\right)^{1/2} \quad (10.8)$$

10.1.3 Surface Charge Density

$$n = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right) \quad (10.9)$$

$$n_s = n_i \exp\left(\frac{e(\phi_{fp}+\Delta\phi_s)}{kT}\right) = n_i \exp\left(\frac{\phi_{fp}+\Delta\phi_s}{V_t}\right) \quad (10.10a)$$

$$or \quad n_s = n_i \exp\left(\frac{\phi_{fp}}{V_t}\right)\cdot\exp\left(\frac{\Delta\phi_s}{V_t}\right)$$

$$,where\Delta\phi_s,is\ the\ surface\ potential\ than\ 2\phi_{fp}$$

$$n_{st} = n_i \exp\left(\frac{\phi_{fp}}{V_t}\right) \quad (10.11)$$

$$n_{st} = n_{st} \exp\left(\frac{\Delta\phi_s}{V_t}\right) \quad (10.12)$$

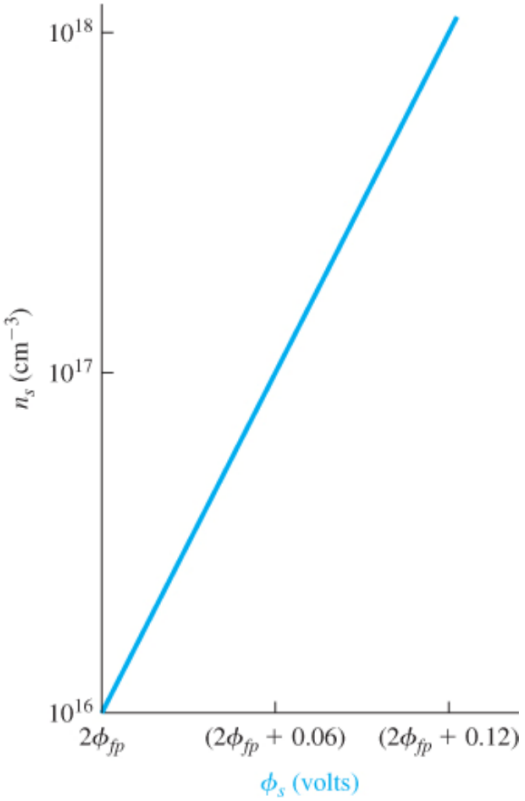


Figure 10.12 | Electron inversion charge density as a function of surface potential.

10.1.4 Work Function Differences

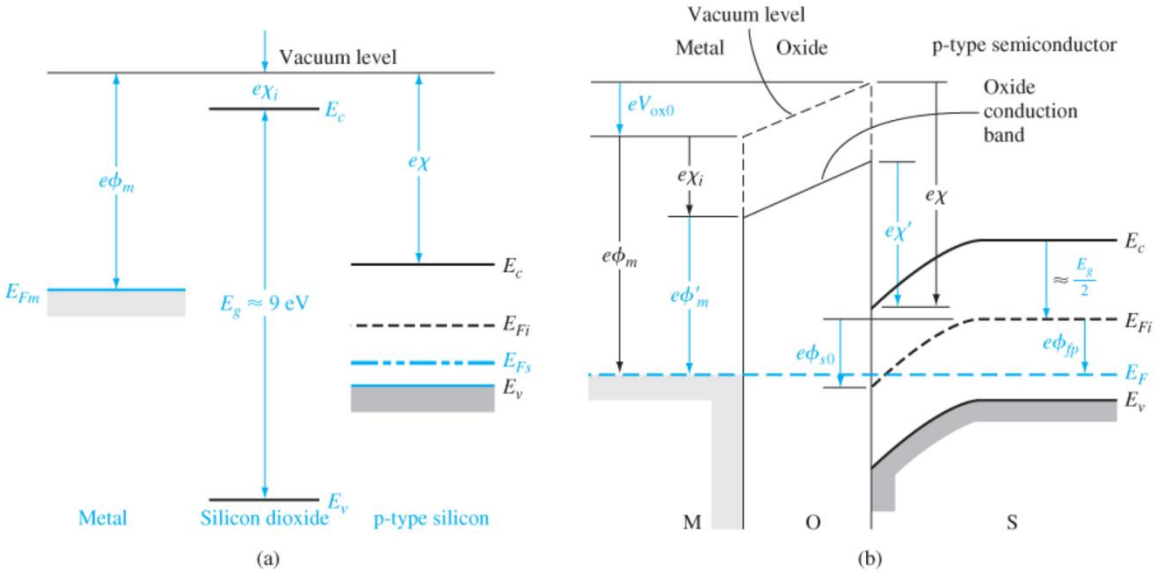


Figure 10.13 | (a) Energy levels in a MOS system prior to contact and (b) energy-band diagram through the MOS structure in thermal equilibrium after contact.

$$e\varphi'_m + eV_{ox0} = e\chi' + \frac{E_g}{2} - e\varphi_{s0} + e\varphi_{fp} \qquad (10.13)$$

$$V_{ox0} + \varphi_{s0} = -\left[\varphi'_m - \left(\chi' + \frac{E_g}{2e} + \varphi_{fp}\right)\right] \qquad (10.14)$$

$$\varphi_{ms} = \left[\varphi'_m - \left(\chi' + \frac{E_g}{2e} + \varphi_{fp}\right)\right] \qquad (10.15)$$

Example 10.2

① **Determine the metal-semiconductor work function difference, φ_{ms}, for a given MOS system and semiconductor doping**

- **Assume the following transistor parameter : φ'_m=3.20V, χ'=3.25V, E_g=1.12V, $N_a=10^{15}$ cm^{-3}**

For silicon at $T = 300K$, *we* may calculate ϕ_{fp} as

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right) = (0.0259)\ln\left(\frac{10^{15}}{1.5\times10^{10}}\right) = 0.288V$$

Then the metal-semiconductor work function difference is

$$\phi_{ms} = \phi_m' - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right) = 3.20 - (3.25 + 0.560 + 0.288), or\ \phi_{ms} = -0.898V$$

- **The value of φ_{ms} will become more negative as the doping of the p-type substrate increases.**

10.1.4 Work Function Differences

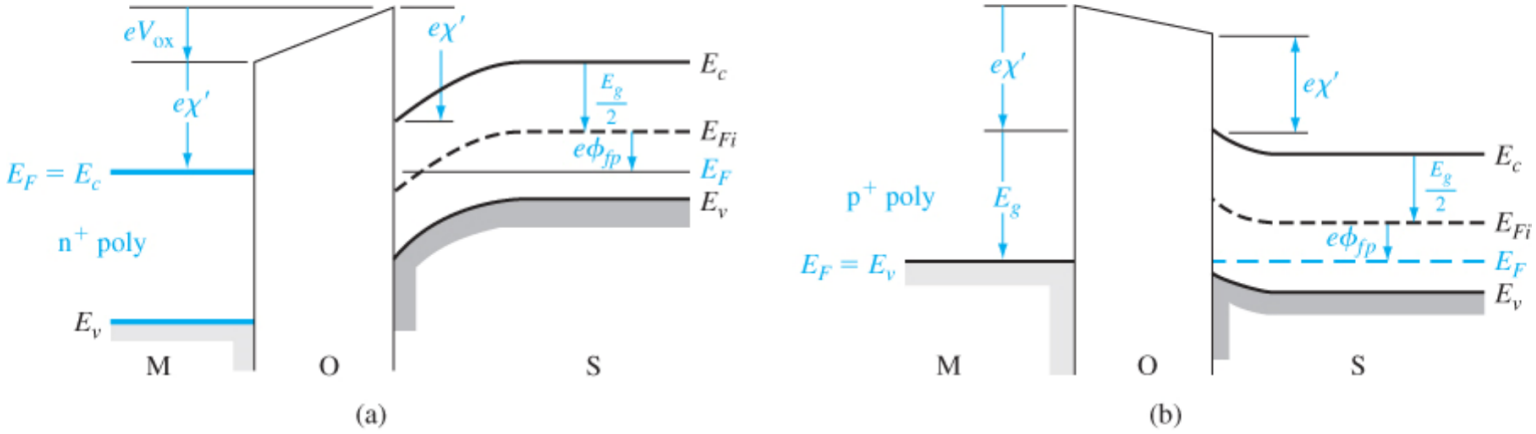


Figure 10.14 | Energy-band diagram through the MOS structure with a p-type substrate at zero gate bias for (a) an n^+ polysilicon gate and (b) a p^+ polysilicon gate.

$$For\ the\ n^+\ polysilicon\ gate,\ \phi_{ms} = [\chi' - (\chi' + \frac{E_g}{2e} + \phi_{fp})] = -\left(\frac{E_g}{2e} + \phi_{fp}\right) \quad (10.16)$$

$$For\ the\ p^+\ polysilicon\ gate,\ \phi_{ms} = [(\chi' + \frac{E_g}{e}) - (\chi' + \frac{E_g}{2e} + \phi_{fp})] = \left(\frac{E_g}{2e} - \phi_{fp}\right) \quad (10.17)$$

10.1.4 Work Function Differences

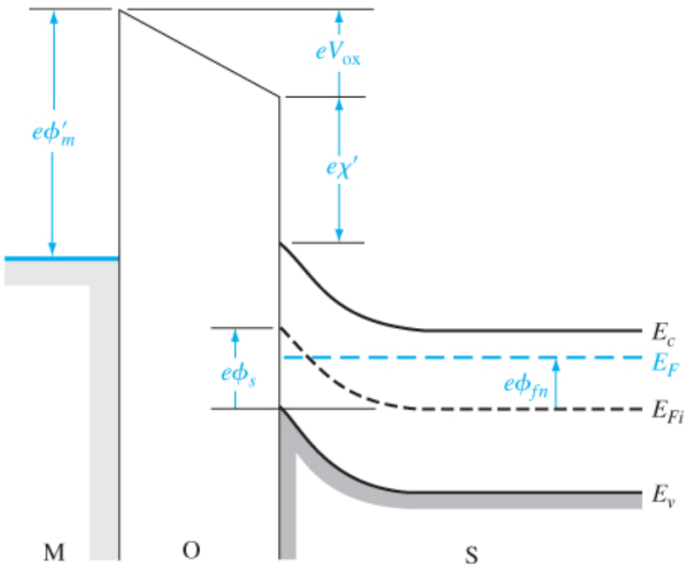


Figure 10.15 | Energy-band diagram through the MOS structure with an n-type substrate for a negative applied gate bias.

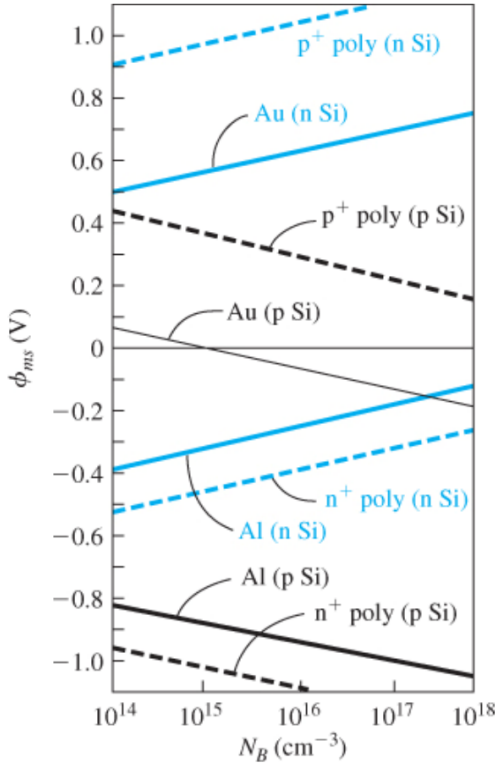


Figure 10.16 | Metal–semiconductor work function difference versus doping for aluminum, gold, and n^- and p^- polysilicon gates.
(*From Sze [17] and Werner [20].*)

$$\phi_{ms} = \phi_m' - \left(\chi' + \frac{E_g}{2e} - \phi_{fn}\right) \quad (10.18)$$

MOS fundamentals 2

10.1.5 Flat-Band Voltage

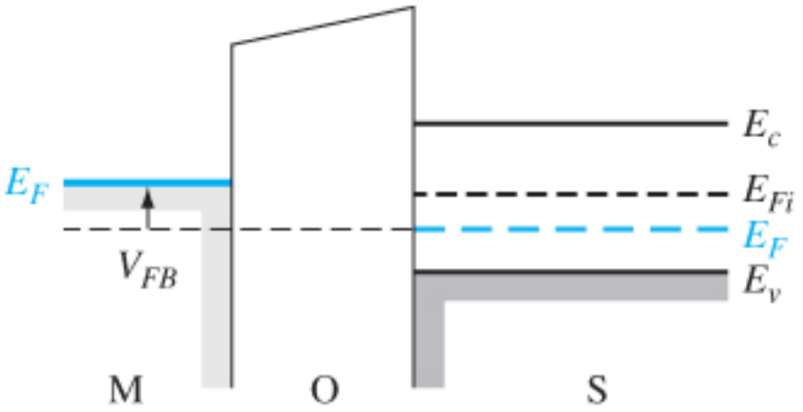


Figure 10.17 | Energy-band diagram of a MOS capacitor at flat band.

- Flat-band voltage is defined as the applied gate voltage;
  - (No band bending) in the semiconductor
  - (Zero net space charge) in this region.
- Work function difference & possible trapped charge in oxide
  - The voltage across the oxide ➔ not zero!!

10.1.5 Flat-Band Voltage (Oxide Charge)

- Assumption
  - There is zero net charge density in the oxide material
- Real situation
  - Net fixed charge density ➔" positive " in the insulator
  - Positive charge
    - ✓ Broken or dangling covalent bonds near the oxide-semiconductor interface.
- Origin of positive charge
  - Thermal oxidation (formation of SiO_2)
    - ✓ Oxygen diffuse through oxide
    - ✓ Reacts near the Si-SiO_2 interface to form the SiO_2.
    - ✓ Si atoms may break away from the silicon material
    - ✓ Oxidation process termination➔ excess Si may exist in the oxide near the interface.
- Strategy for oxide charge.
  - ✓ Annealing the oxide in an argon or nitrogen atmosphere
  - ✓ However, the charge is rarely zero.

10.1.5 Oxide Charge Distribution

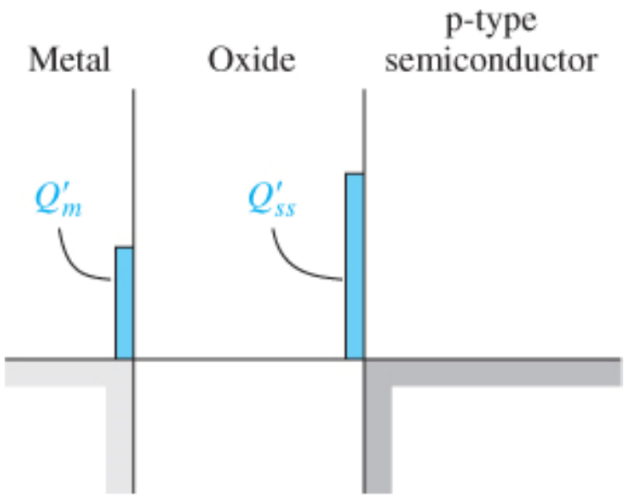


Figure 10.18 | Charge distribution in a MOS capacitor at flat band.

- The net fixed charge in the oxide
  - Located fairly close to the oxide semiconductor-interface
  - Equivalent trapped charge per unit area (Q_{ss}')
    - ✓ Located in the oxide directly adjacent to the oxide-semiconductor interface.

10.1.5 Flat Band Voltage Derivation (1)

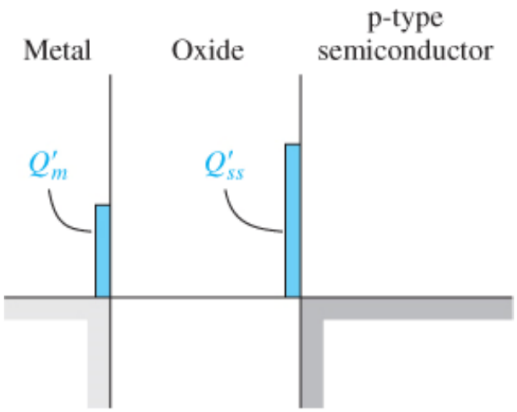


Figure 10.18 | Charge distribution in a MOS capacitor at flat band.

- Zero applied gate voltage.

$$V_{ox0} + \phi_{s0} = -\phi_{ms} \quad (10.19)$$

- Gate voltage ➔(potential drop) across the oxide and (the surface potential) ➔ change!

$$V_G = \Delta V_{ox} + \Delta\phi_s = (V_{ox} - V_{ox0}) + (\phi_s - \phi_{s0}) \quad (10.20)$$

- Using Equation (10.19)

$$V_G = \Delta V_{ox} + \phi_s + \phi_{ms} \quad (10.21)$$

- Zero net charge in the semiconductor
- Equivalent fixed surface charge density exist in the oxide.

➔ The charge density on the metal is Q_m' and from charge neutrality.

$$Q'_m + Q'_{ss} = 0 \quad (10.22)$$

10.1.5 Flat Band Voltage Derivation (2)

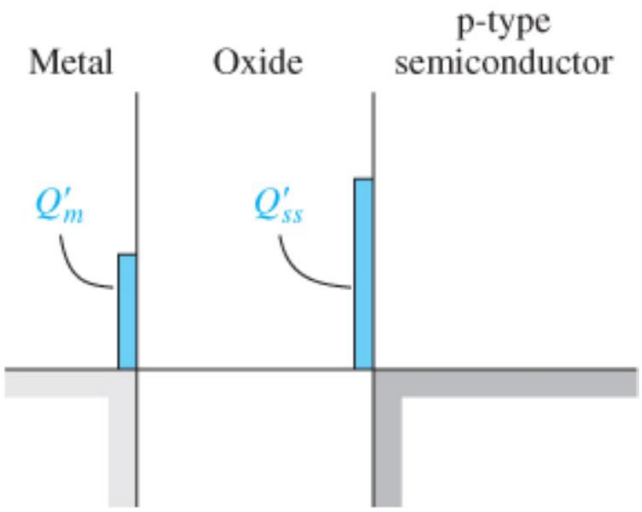


Figure 10.18 | Charge distribution in a MOS capacitor at flat band.

$$Q'_m + Q'_{ss} = 0 \quad (10.22)$$

- Q_m' to the voltage across the oxide

$$V_{ox} = \frac{Q'_m}{C_{ox}} \quad (10.23)$$

- C_{ox} is the oxide capacitance per unit area.
- Substitution: Equ. (10.22)➔ Equ. (10.23)

$$V_{ox} = \frac{-Q'_{ss}}{C_{ox}} \quad (10.24)$$

- In the flat-band condition,
  - ✓ the surface potential is zero (or ϕ_s=0)

- Flat-band voltage for MOS device:

$$V_G = V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}} \quad (10.25)$$

Example 10.3

① **Calculate the flat-band voltage for a MOS capacitor with a p-type semiconductor substrate**

- **Assume the following transistor parameter : $N_a=10^{16}$ cm^{-3}, t_{ox}=20nm=200Å, $Q'_{ss}=5\times10^{10}$ electronic charges per cm^2**

The work function difference, from Figure 10.16, is $\phi_{ms} \cong -1.1V$. The oxide capacitance is found to be

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.85\times10^{-14})}{200\times10^{-8}} = 1.726\times10^{-7}\, F/cm^2$$

The equivalent oxide charge density is

$$Q'_{SS} = (5\times10^{10})(1.6\times10^{-19}) = 8\times10^{-9}\, C/cm^2$$

The flat-band voltage is then determined to be

$$V_{FB} = \phi_{ms} - \frac{Q'_{SS}}{C_{ox}} = -1.1 - \frac{8\times10^{-9}}{1.726\times10^{-7}} = -1.15V$$

- **The applied gate voltage required to achieve the flat-band condition for this p-type substrate is negative. If the amount of fixed oxide charge increase, the flat-band voltage becomes even more negative.**

10.1.6 Threshold Voltage (1)

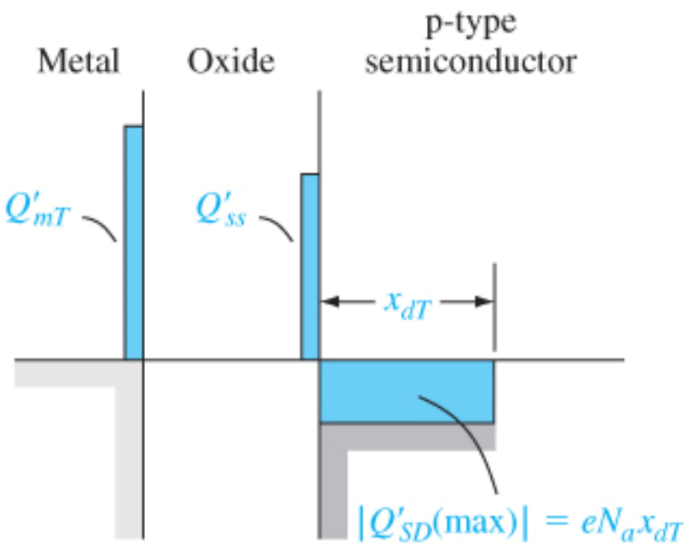


Figure 10.19 | Charge distribution in a MOS capacitor with a p-type substrate at the threshold inversion point.

- Applied gate voltage
  - ✓ To achieve the threshold inversion point
  - ✓ The surface potential is
    - $\phi_s = 2\phi_{fp}$ for p-type Si
    - $\phi_s = 2\phi_{fn}$ for n-type Si

- Charge distribution through the MOS device.
  - ✓ The space charge width➔ maximum
- Assumption
    - Inversion charge is neglected.
- Charge conservation law

$$Q'_{mT} + Q'_{ss} = \left|Q'_{SD}(\max)\right| \quad (10.26)$$

Where $\left|Q'_{SD}(\max)\right| = eN_a x_{dT}$ (10.27)

10.1.6 Threshold Voltage (2)

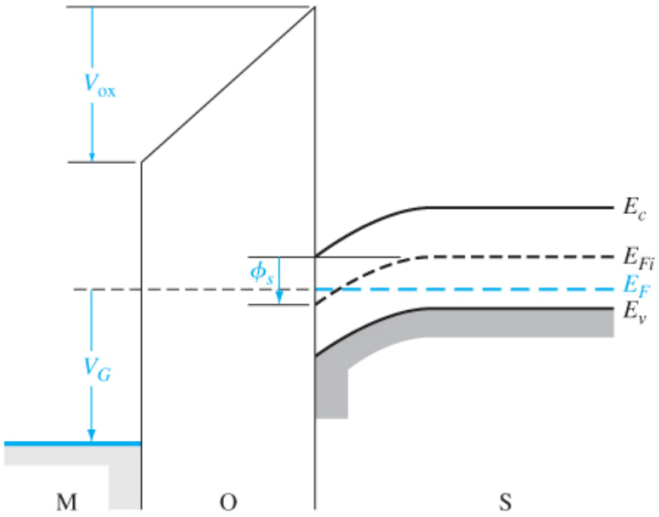


Figure 10.20 | Energy-band diagram through the MOS structure with a positive applied gate bias.

- The energy-band diagram of the MOS system
  - ✓ With applied positive voltage
  - ✓ Change the voltage of Vox
  - ✓ Surface potential

$$V_G = \Delta V_{ox} + \Delta\phi_s = (V_{ox} - V_{ox0}) + (\phi_s - \phi_{s0}) \quad (10.20)$$

- At threshold, we can $V_G=V_{TN}$.
  - ✓ VTN is the threshold voltage
  - ✓ Electron inversion layer charge
- The surface potential at $V_G=V_{TN}$
  - $\phi_s=2\phi_{fp}$

$$V_{TN} = V_{oxT} + 2\phi_{fp} + \phi_{ms} \quad (10.28)$$

- V_{oxT} (voltage across the oxide at $V_G=V_{TN}$)

$$V_{oxT} = \frac{Q'_{mT}}{C_{ox}} \quad (10.29)$$

10.1.6 Threshold Voltage (3)

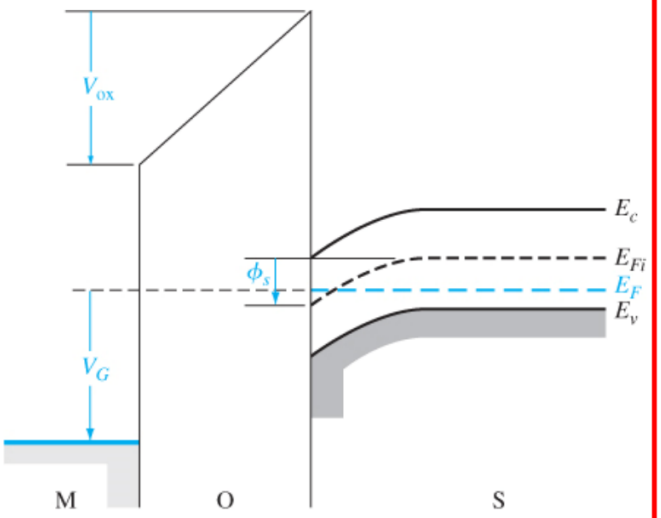


Figure 10.20 | Energy-band diagram through the MOS structure with a positive applied gate bias.

- By using Equation (10.26)

$$Q'_{mT} + Q'_{ss} = \left|Q'_{SD}(\max)\right| \quad (10.26)$$

- We can write:

$$V_{oxT} = \frac{Q'_{mT}}{C_{ox}} = \frac{1}{C_{ox}}\left(\left|Q'_{SD}(\max)\right| - Q'_{ss}\right) \quad (10.30)$$

- Finally, threshold voltage can be written:

$$V_{TN} = \frac{\left|Q'_{SD}(\max)\right|}{C_{ox}} - \frac{Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp} \quad (10.31a)$$

$$V_{TN} = \left(\left|Q'_{SD}(\max)\right| - Q'_{ss}\right)\left(\frac{t_{ox}}{\varepsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fp} \quad (10.31b)$$

- Using " flat-band voltage from Eq. (10.25)

$$V_{TN} = \frac{\left|Q'_{SD}(\max)\right|}{C_{ox}} + V_{FB} + 2\phi_{fp} \quad (10.31c)$$

Example 10.4(1)

① **Calculate the threshold voltage of a MOS system using an aluminum gate.**

- **Assume the following transistor parameter : T=300K, $N_a=10^{15}$ cm^{-3}, $Q'_{ss}=10^{10}$ cm^{-2}, t_{ox}=12nm=120Å**

From Figure 10.16, we find $\phi_{ms} \cong -0.88V$. The other parameters are

$$\varphi_{fp} = V_t \ln\left(\frac{N_a}{ni}\right) = (0.0259)\ln\left(\frac{10^{15}}{1.5\times10^{10}}\right) = 0.2877V$$

and

$$\chi_{dT} = \left\{\frac{4\varepsilon_s\phi_{fp}}{eN_a}\right\}^{1/2} = \left\{\frac{4(11.7)(8.85\times10^{-14})(0.2877)}{(1.6\times10^{-19})(10^{15})}\right\}^{1/2} = 8.63\times10^{-5}cm$$

Example 10.4(2)

① **Calculate the threshold voltage of a MOS system using an aluminum gate.**

- **Assume the following transistor parameter : T=300K, $N_a=10^{15}$ cm^{-3}, $Q'_{ss}=10^{10}$ cm^{-2}, t_{ox}=12nm=120Å**

Then

$$|Q'_{SD}(\max)| = eN_a x_{dT} = (1.6\times10^{-19})(10^{15})(8.63\times10^{-5}) = 1.381\times10^{-8}\,C/cm^2$$

The threshold voltage is now found to be

$$V_{TN} = (|Q'_{SD}(\max)| - Q'_{SS})\left(\frac{t_{ox}}{\varepsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fp}$$

$$= \left[(1.381\times10^{-8}) - (10^{10})(1.6\times10^{-19})\right]\bullet\left[\frac{120\times10^{-8}}{(3.9)(8.85\times10^{-14})}\right] + (-0.88) + 2(0.2877)$$

$$V_{TN} = -0.262V$$

- **In this example, the semiconductor is fairly lightly doped, which, in conjunction with the positive charge in the oxide and the work function difference, is sufficient to induce an electron inversion layer charge even with zero applied gate voltage. This condition makes the threshold voltage negative.**

10.1.6 Threshold Voltage (4)

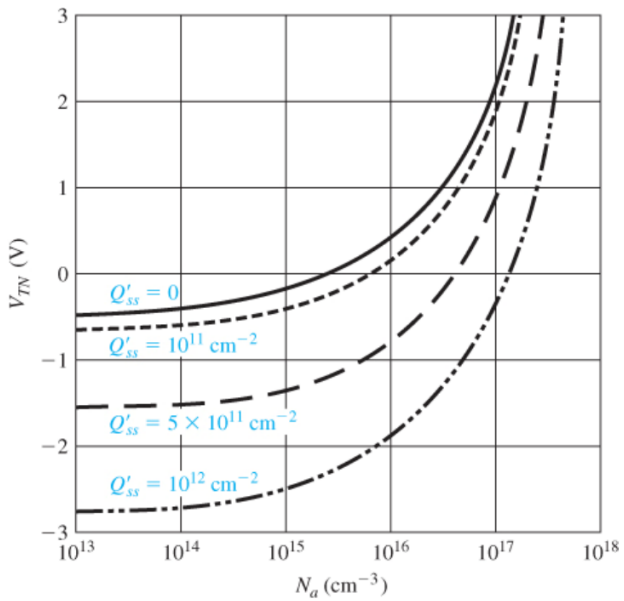


Figure 10.21 | Threshold voltage of an n-channel MOSFET versus the p-type substrate doping concentration for various values of oxide trapped charge (t_{ox} = 500 Å, aluminum gate).

- Negative Vth for p-type substrate
  - ✓ Depletion mode device
  - ✓ Hole carrier already exist➔ channel is formed even at VGS=0V.
- P-type semiconductor should be heavily
  - ✓ Operation for enhancement mode

10.1.6 Threshold Voltage-n-type Si (5)

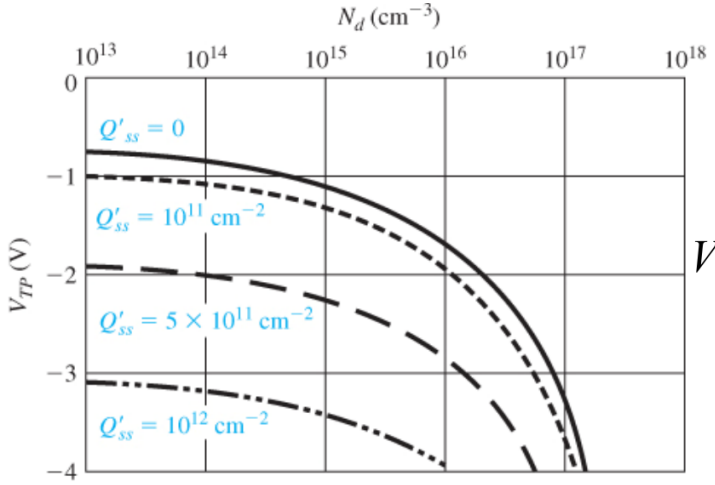


Figure 10.22 | Threshold voltage of a p-channel MOSFET versus the n-type substrate doping concentration for various values of oxide trapped charge (t_{ox} = 500 Å, aluminum gate).

- The same concept derivation
  - ✓ Negative gate voltage
  - ✓ Inversion of holes at the oxide-semiconductor interface.

$$V_{TP} = \left(-\left|Q'_{SD}\left(\max\right)\right| - Q'_{ss}\right)\left(\frac{t_{ox}}{\varepsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fn} \qquad (10.32)$$

- Where

$$\phi_{ms} = \phi'_m - \left(\chi' + \frac{E_g}{2e} - \phi_{fn}\right) \qquad (10.33a)$$

$$\left|Q'_{SD}\left(\max\right)\right| = eN_d x_{dT} \qquad (10.33b)$$

$$x_{dT} = \left\{\frac{4\varepsilon_s \phi_{fn}}{eN_d}\right\}^{1/2} \qquad (10.33c)$$

$$\phi_{fn} = V_t \ln\left(\frac{N_d}{n_i}\right) \qquad (10.33d)$$

- Note:
  - ✓ X_{dT} and ϕ_{fn} is positive
  - ✓ VTP (threshold voltage)
    - ✓ Inversion of holes
  - ✓ N and P subscript will be drop.

Example 10.5(1)

① **Determine the gate material and design the semiconductor doping concentration to yield a specified threshold voltage.**

- **Assume the following transistor parameter : t_{ox}=12nm=120Å, Q'_{ss}=2×10^{10} cm^{-2}, V_{TP}=−0.1V**

Consider a doping concentration of $N_d = 10^{17}\,cm^{-3}$.

From Figure 10.16, the metal - semiconductor work function difference is $\phi_{ms} \cong +1.1\text{V}$.

The remaining parameters are found to be

$$\phi_{fn} = V_t \ln\left(\frac{N_d}{n_i}\right) = (0.0259)\ln\left(\frac{10^{17}}{1.5\times10^{10}}\right) = 0.407V$$

$$\chi_{dT} = \left(\frac{4\varepsilon_s \phi_{fn}}{eN_d}\right)^{1/2} = \left\{\frac{4(11.7)(8.85\times10^{-14})(0.407)}{(1.6\times10^{-19})(10^{17})}\right\}^{1/2} = 1.026\times10^{-5}\,cm$$

$$\left|Q'_{SD}(\max)\right| = eN_d\chi_{dT} = (1.6\times10^{-19})(10^{17})(1.026\times10^{-5}) = 1.642\times10^{-7}\,C/cm^2$$

- **The current capability of a MOSFET is directly proportional to the channel width W. The current handling capability can be increased by increasing W.**

Example 10.5(2)

① **Determine the gate material and design the semiconductor doping concentration to yield a specified threshold voltage.**

- **Assume the following transistor parameter : t_{ox}=12nm=120Å, Q'_{ss}=2×10^{10} cm^{-2}, V_{TP}=−0.1V**

The threshold voltage is determined to be

$$V_{TP} = \left[-\left|Q'_{SD}(\max)\right| - Q'_{SS}\right]\cdot\left(\frac{t_{ox}}{\varepsilon_{ox}}\right) + \phi_{ms} - 2\phi_{fn}$$

or

$$V_{TP} = \frac{\left[-(1.642\times10^{-7}) - (2\times10^{10})(1.6\times10^{-19})\right]\cdot(120\times10^{-8})}{(3.9)(8.85\times10^{-14})} + 1.1 - 2(0.407)$$

which yields

$$V_{TP} = -0.296V \cong -0.3V$$

- **The negative threshold voltage, with the n-type substrate, implies that this MOS capacitor is an enhancement mode device. The inversion layer charge is zero with zero applied gate voltage, and a negative gate voltage must be applied to induce the hole inversion.**

10.2 Capacitance-Voltage Characteristics

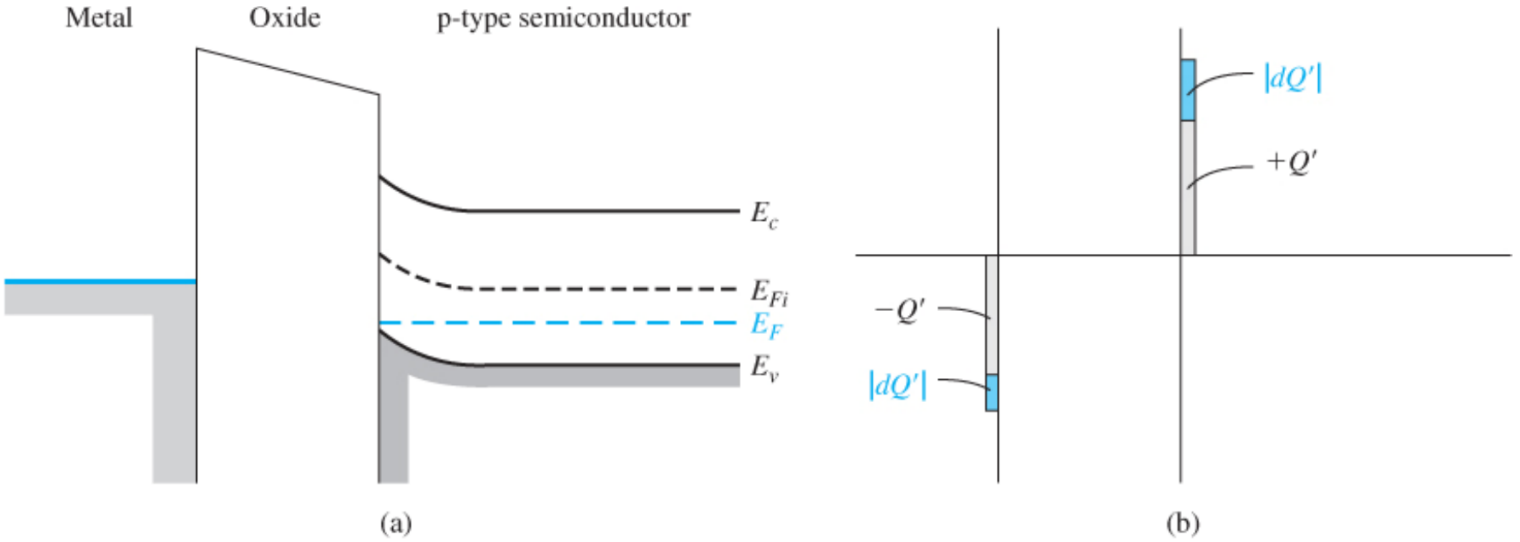


Figure 10.23 | (a) Energy-band diagram through a MOS capacitor for the accumulation mode. (b) Differential charge distribution at accumulation for a differential change in gate voltage.

- What is capacitance??

$$C = \frac{dQ}{dV} \quad (10.34)$$

  - ✓ dQ: differential change in charge on one plate
  - ✓ dV: differential change in voltage dV across the capacitor

10.2 Capacitance-Voltage Characteristics

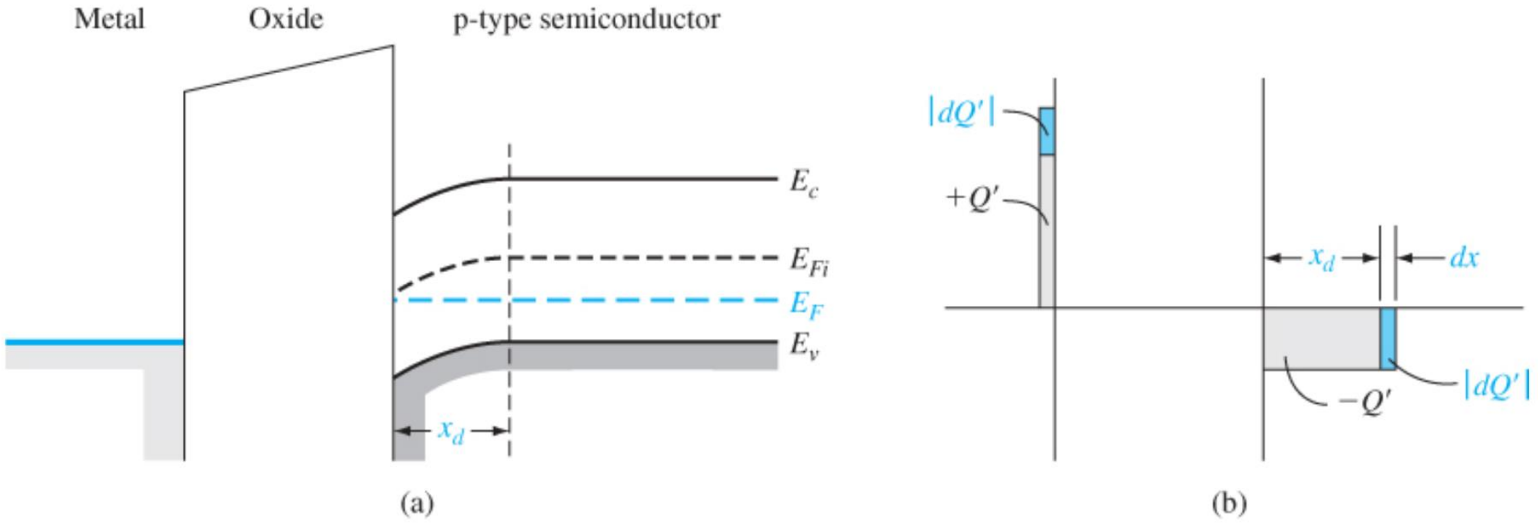


Figure 10.24 | (a) Energy-band diagram through a MOS capacitor for the depletion mode. (b) Differential charge distribution at depletion for a differential change in gate voltage.

$$C'(acc) = C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \quad (10.35)$$

10.2 Capacitance-Voltage Characteristics(1)

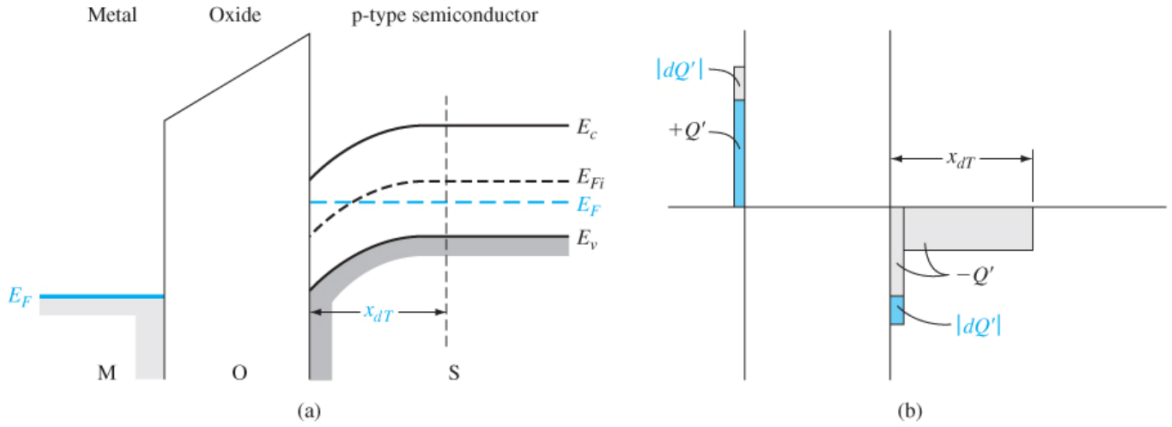


Figure 10.25 | (a) Energy-band diagram through a MOS capacitor for the inversion mode. (b) Differential charge distribution at inversion for a low-frequency differential change in gate voltage.

$$\frac{1}{C'(depl)} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}} \qquad (10.36a)$$

$$C'(depl) = \frac{C'_{ox}\,C'_{SD}}{C_{ox} + C'_{SD}} \qquad (10.36b)$$

$$C'(depl) = \frac{C_{ox}}{1 + \dfrac{C_{ox}}{C'_{SD}}} = \frac{\varepsilon_{ox}}{t_{ox} + \left(\dfrac{\varepsilon_{ox}}{\varepsilon_s}\right)x_d} \qquad (10.37)$$

$$C'_{\min} = \frac{\varepsilon_{ox}}{t_{ox} + \left(\dfrac{\varepsilon_{ox}}{\varepsilon_s}\right)x_{dT}} \qquad (10.38)$$

- As the space charge width increases, the total capacitance decreases.

10.2 Capacitance-Voltage Characteristics(2)

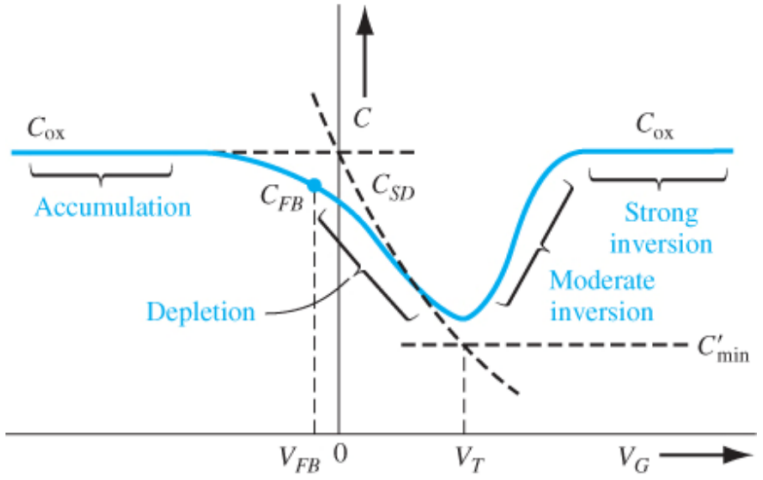


Figure 10.26 | Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate. Individual capacitance components are also shown.

$$C'_{FB} = \frac{\varepsilon_{ox}}{t_{ox} + \left(\frac{\varepsilon_{ox}}{\varepsilon_s}\right)\sqrt{\left(\frac{kT}{e}\right)\left(\frac{\varepsilon_s}{eN_a}\right)}} \qquad (10.40)$$

- Three dashed segments
  - ✓ C_{ox}
  - ✓ C_{SD}
  - ✓ C_{min}
- The flat band capacitance: function of oxide thickness and semiconducting doping

Example 10.6(1)

① **Calculate C_{ox}, C'_{min} and C'_{FB} for a MOS capacitor.**

- **Assume the following transistor parameter : T=300K, $N_a=10^{16}$ cm^{-3}, t_{ox}=18nm=180Å**

The oxide capacitance is

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.85\times10^{-14})}{180\times10^{-8}} = 1.9175\times10^{-7}\, F/cm^2$$

To find the minimum capacitance, we need to calculate

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right) = (0.0259)\ln\left(\frac{10^{16}}{1.5\times10^{10}}\right) = 0.3473V$$

and

$$\chi_{dT} = \left\{\frac{4\varepsilon_s\phi_{fp}}{eN_d}\right\}^{1/2} = \left\{\frac{4(11.7)(8.85\times10^{-14})(0.3473)}{(1.6\times10^{-19})(10^{16})}\right\}^{1/2} \cong 0.30\times10^{-4}\, cm$$

Then

$$C'_{\min} = \frac{\varepsilon_{ox}}{t_{ox} + \left(\frac{\varepsilon_{ox}}{\varepsilon_s}\right)\chi_{dT}} = \frac{(3.9)(8.85\times10^{-14})}{180\times10^{-8} + \left(\frac{3.9}{11.7}\right)(0.30\times10^{-4})} = 2.925\times10^{-8}\, F/cm^2$$

Example 10.6(2)

① **Calculate C_{ox}, C'_{min} and C'_{FB} for a MOS capacitor.**

- **Assume the following transistor parameter : T=300K, $N_a=10^{16}$ cm^{-3}, t_{ox}=18nm=180Å**

We may note that

$$\frac{C'_{\min}}{C_{ox}} = \frac{2.925\times10^{-8}}{1.9175\times10^{-7}} = 0.1525$$

The flat - band capacitance is

$$C'_{FB} = \frac{\varepsilon_{ox}}{t_{ox} + \left(\frac{\varepsilon_{ox}}{\varepsilon_s}\right)\sqrt{\frac{V_t\varepsilon_s}{eN_a}}} = \frac{(3.9)(8.85\times10^{-14})}{180\times10^{-8} + \left(\frac{3.9}{11.7}\right)\sqrt{\frac{(0.0259)(11.7)(8.85\times10^{-14})}{(1.6\times10^{-19})(10^{16})}}} = 1.091\times10^{-7}\ F/cm^2$$

We also note that

$$\frac{C'_{FB}}{C_{ox}} = \frac{1.091\times10^{-7}}{1.9175\times10^{-7}} = 0.569$$

- **The ratios of C'_{min}/C_{ox} and C'_{FB}/C_{ox} are typical values obtained in C-V plots.**

10.2 Capacitance-Voltage Characteristics(3)

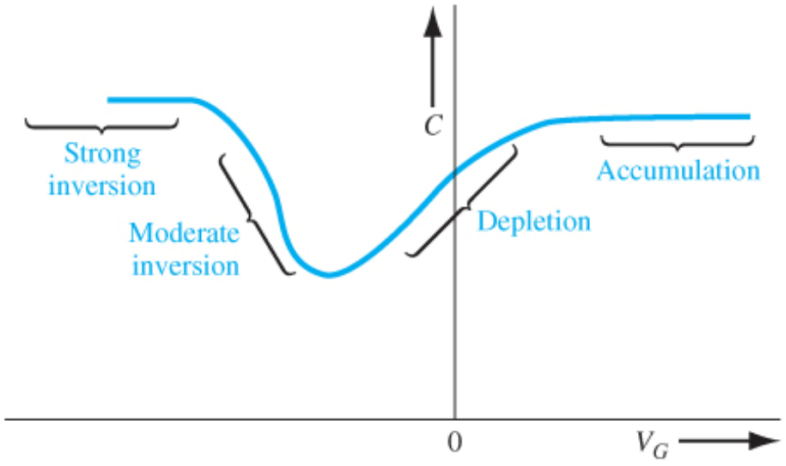


Figure 10.27 | Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with an n-type substrate.

- Assumption
  - ✓ C_{ox}=1.9175 x 10^{-7} F/cm^2
  - ✓ L=2 μm, W= 20 μm
  - ✓ C_{oxT}=C_{ox}LW=(1.9175 x 10^{-7} F/cm^2) x (2 x 10^{-4} cm) x (20 x 10^{-4} cm)
    = 7.67 x 10^{-14} F=76.7 fF
- Accumulation condition can be obtained at positive bias condition.

10.2.2 Frequency Effects

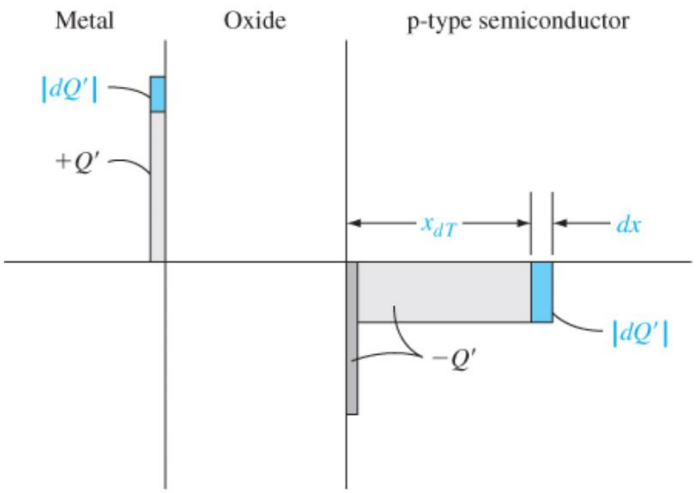


Figure 10.28 | Differential charge distribution at inversion for a high-frequency differential change in gate voltage.

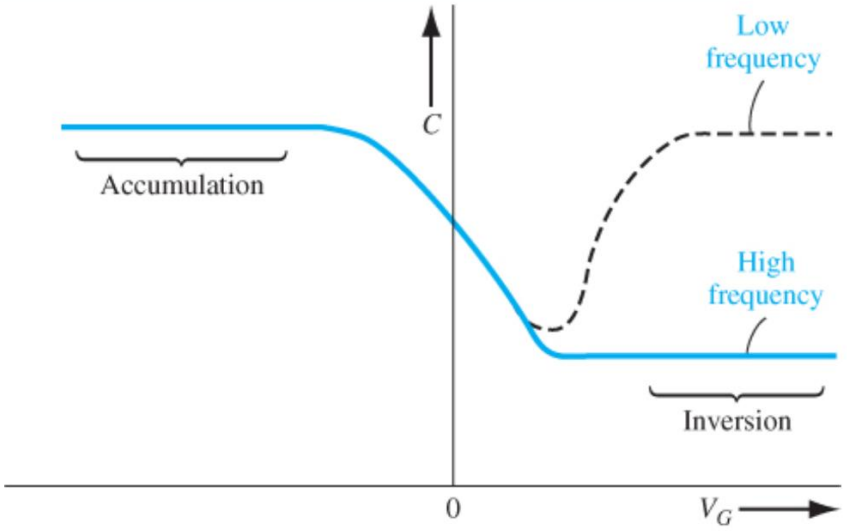


Figure 10.29 | Low-frequency and high-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate.

- Assumption
  - ✓ C_{ox}=1.9175 x 10^{-7} F/cm^2

10.2.3 Fixed Oxide and Interface Charge Effects

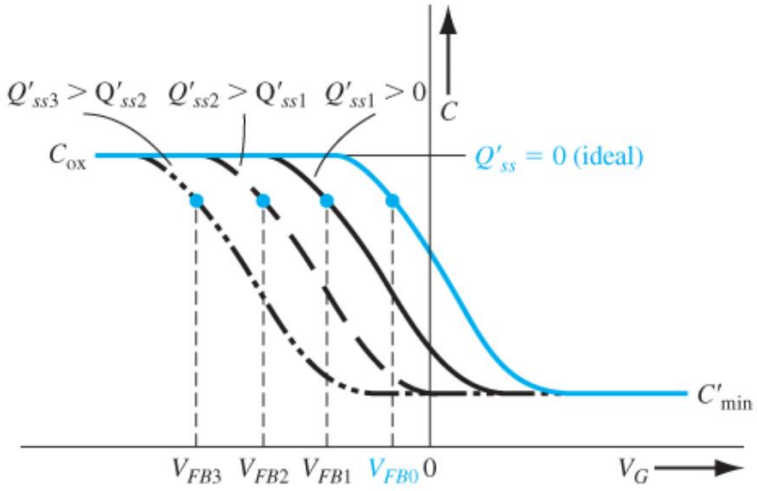


Figure 10.30 | High-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate for several values of effective trapped oxide charge.

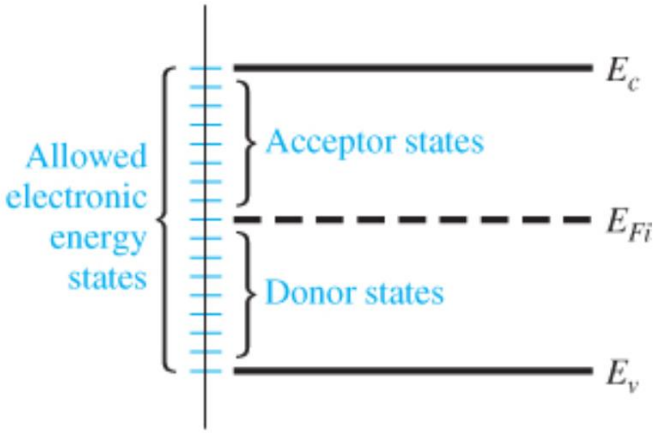


Figure 10.31 | Schematic diagram showing interface states at the oxide–semiconductor interface.

- Assumption
  - ✓ C_{ox}=1.9175 x 10^{-7} F/cm^2

10.2.3 Fixed Oxide and Interface Charge Effects (2)

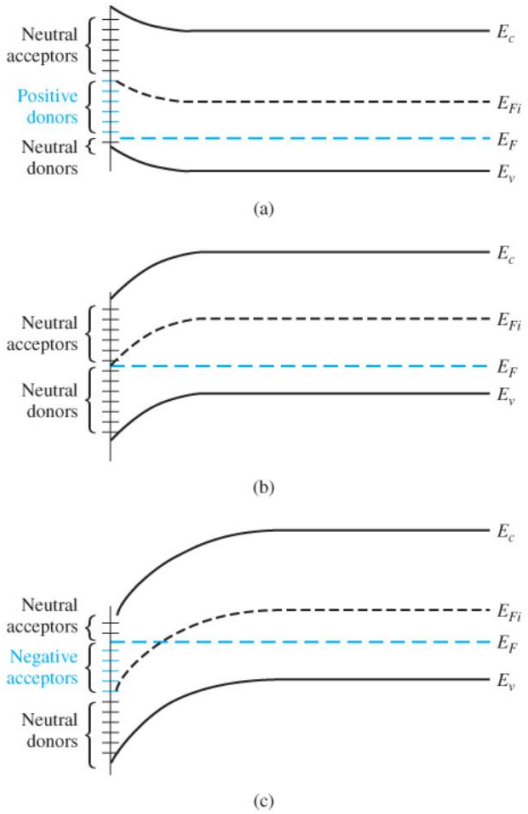


Figure 10.32 | Energy-band diagram in a p-type semiconductor showing the charge trapped in the interface states when the MOS capacitor is biased (a) in accumulation, (b) at midgap, and (c) at inversion.

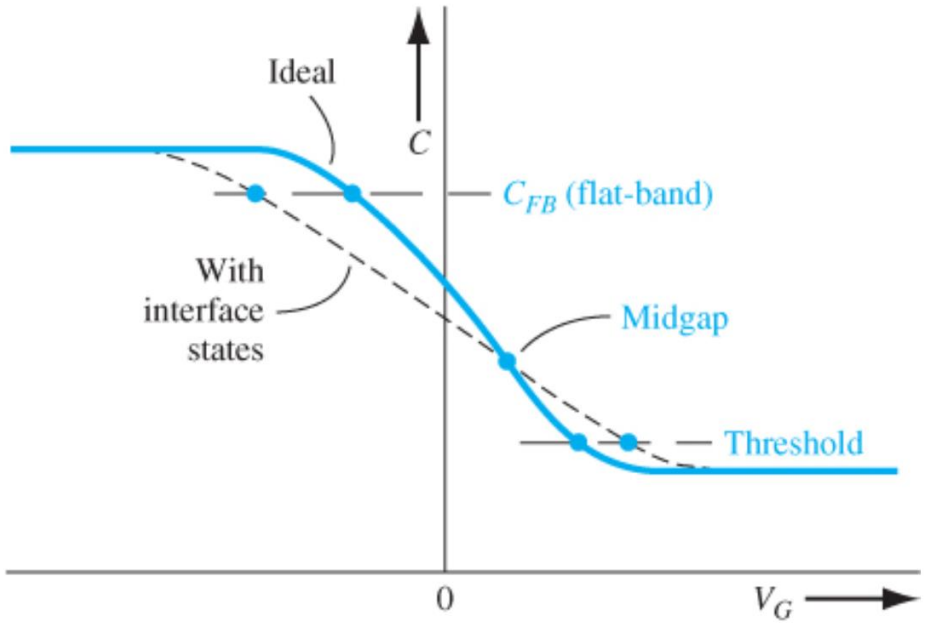


Figure 10.33 | High-frequency C–V characteristics of a MOS capacitor showing effects of interface states.